Maximilian Purcea

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Education

Cornell University, College of Engineering — Ithaca, NY

Bachelor of Science in Mechanical Engineering — *Expected May 2028*

- *Relevant Coursework:* Fluid Mechanics, Thermodynamics, Mechanical Design, Mechanics of Engineering Materials, Statics and Mechanics of Solids, Dynamics, System Dynamics, Differential Equations for Engineers

Engineering Experience

Princeton Carbon Works — June 2026 – August 2026

Engineering Intern — *Mystic, CT*

- Modeled 75 and 95 mm triathlon rim profiles in SolidWorks and Fusion 360, defining each cross-section by four geometric parameters, and ran full yaw-range CFD sweeps on every candidate in NablaFlow AeroCloud.
- Wrote Python scripts that weight drag across eight yaw bins by the time a rider spends at each angle; found maximum rim width to be the dominant parameter, with an optimum near 115% of tire width at both depths.
- Produced SolidWorks manufacturing drawings for a carbon frame with GD&T callouts, and reverse-engineered Shimano and SRAM UDH derailleur hangers and 3D-printed replicas that fit a prototype bike.
- Built a Python and Excel tool that reads live BOM data, flags builds short on parts, and assigns each shortfall to one of two contract factories, replacing a recurring manual review.

Cornell Concrete Canoe Project Team — September 2025 – Present

Mold Sub-Team Engineer — *Ithaca, NY*

- Model hull and mold geometry in Fusion 360 for a 20 ft competition canoe and run hull CFD in ANSYS Fluent to compare design revisions.
- Rework the foam mold process so a single mold holds up over multiple pours, tightening casting tolerances and cutting the patchwork needed between pours.
- Turn mix sub-team test results and shop machining limits into specific hull and mold geometry revisions.

AfterQuery — September 2026 – Present

Project Administrator (Contract) — *Remote*

- Administer an expert data project building visual reasoning benchmarks for frontier AI models.
- Review, revise, and approve expert submissions against quality, difficulty, and reproducibility standards.
- Analyze model outputs to trace the reasoning path behind each AI conclusion, and refine submissions to reflect the steps models take.

Handshake AI — October 2025 – Present

AI Research & Evaluation Specialist — *Remote*

- Write multi-step, search-dependent research questions and document the source path behind each answer, with every supporting fact drawn from an independent source.
- Grade model responses for factual accuracy and reasoning across text, image, audio, and video inputs; flag recurring failure modes to inform training-data curation.

Leadership & Additional Experience

Cornell Lightweight Crew — August 2024 – Present

IRA Division I Varsity Athlete — *Ithaca, NY*

- Train and compete 20+ hours per week in a nationally ranked Division I program while carrying a full engineering course load.
- Rowed in the crew that set the all-time course record at the Princeton Chase Regatta.

Pelham Country Club — Summers 2024 & 2025

Head Counselor — *Pelham, NY*

- Ran daily operations for a summer camp of 50+ members and supervised a staff of junior counselors.
- Built weekly staff schedules, covered call-outs, and handled member and guardian concerns.

Technical Skills & Additional Information

CAD & Simulation: SolidWorks, Fusion 360, AutoCAD, ANSYS Fluent, CFD (NablaFlow AeroCloud), GD&T
Programming & Analysis: Python (NumPy, pandas, Matplotlib), C, Excel, LaTeX, MATLAB
Fabrication & Testing: Carbon-fiber layup, foam mold fabrication, 3D printing, prototype assembly and testing
Languages: English (native), Romanian (fluent), French (proficient)
Certifications: First Aid/CPR/AED, PADI Advanced Open Water Diver
Interests: Backpacking, skiing, scuba diving, chess, drumming, ultramarathons, triathlons, DJ music mixing/production